Independent reconstruction of the documented coastal adaptation model of Wong et al. (2025)

Research-agent record prepared for Sophia Ludtke

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Abstract

Coastal cost distributions depend on both sea-level response and the joint uncertainty assigned to climate, exposure, and adaptation. This project examines whether the results of Wong et al. (2025) can be reconstructed from scholarly documentation without consulting the authors' implementation or computational outputs. We recover component equations, implement numerical kernels in Julia, and compare manufactured cases with analytical limits and an independent SciPy integration. The calculations verify heat conservation, ice-response convergence, flood-loss integration, and economic accounting. They also expose a finite-step residual in the printed emissions recurrence and a lower-bound discrepancy in an Antarctic temperature transform. These are findings about written equations, not evidence of defects in the inaccessible original software. A complete independent joint calibration and the target coastal input bundle were not recovered. Consequently, none of the thirteen numerical target figures is independently reproduced. An original workflow diagram, verified component calculations, a complete figure register, and explicit recovery requirements provide an auditable basis for further work.

1 Objective and evidentiary boundary

Wong et al. [1] estimate global coastal adaptation costs and losses conditional on warming in 2100. Their model connects emissions, carbon cycling, temperature and ocean heat, global and local sea level, and segment-level coastal decisions. The scientific objective here is to reproduce the resulting conditional distributions from the documented mechanisms. It is not sufficient to draw curves resembling the published figures or to reproduce marginal parameter ranges while discarding their dependence.

The independence restriction permits scientific papers, their supplementary methods, and exogenous inputs from independent providers. Published figures and printed summary statistics are comparison evidence only. No source code, processed input bundle, posterior sample, simulation file, or plotting script from the target authors was opened or used. An initial broad literature search displayed incidental repository-index snippets, including a data-loading excerpt. Those results were discarded, and subsequent searches focused on scientific documents and provider records. This exposure is retained in the research log rather than represented as complete isolation from search results.

The recovered sources include the target article and its ten-figure supplement, the SNEASY climate description [3], its carbon-cycle reference [4], BRICK and MimiBRICK descriptions [5, 6], the DAIS paper [7], and the CIAM mathematical supplement [8]. The December 2024 revision of the cited emissions preprint [2] was recovered through public scholarly file-version metadata.

Its current DOI landing page serves a later 2025 version. Since the target citation does not pin a revision, this historical manuscript remains a version-qualified specification. Stable acquisition URLs and hashes are recorded in `docs/SOURCES.csv`.

2 Original experiment and recovered interfaces

The target source ensemble contains 100,000 states of the world, with 5,000 used for the coastal analysis. Global warming in 2100, relative to 1850–1900, defines four bins: at most 1.5 °C, above 1.5 and at most 2.5 °C, above 2.5 and at most 3.5 °C, and above 3.5 °C. Global sea level in 2100 uses a 1995–2014 reference. The baseline socioeconomic scenario is SSP4; SSP2 provides a sensitivity case. Monetary outcomes are in 2010 US dollars with a 4% annual discount rate [1].

The coastal model uses 12,148 DIVA segments and decadal calculations from 2010 through 2150. Planning intervals are 2010–2050, 2060–2100, and 2110–2150. Protection and retreat standards are associated with return periods of 1, 10,100, 1,000, and 10,000 years, alongside no adaptation. The paper specifies nine adaptation combinations; the excluded combination is not identified by the recovered text. A nominal one-year return period also requires a convention beyond the usual GEV quantile at probability $1 - 1/R$, which is an endpoint when $R = 1$. Neither detail is silently filled in here.

The methods describe both periodic optimization and persistence of an initial strategy and return level, with later height augmentation. Actual adaptation already in place is not globally observed; the original calculation initializes an optimal adaptation state. Reconstructing this spin-up requires the same exposure and extreme-water inputs as the forward model. It cannot be inferred from a global cost total.

Figure1 identifies the implemented kernels and the missing links. These kernels can be exercised independently. They do not constitute a calibrated climate-to-cost simulation.

3 Independent mathematical reconstruction

3.1 Emissions and climate kernels

For annual emissions q_t in $\mathrm{Gt\,CO_2\,yr^{-1}}$, the target's printed equation is

$$q_t = \begin{cases} q_{t-1} + \gamma_g(\tau - t), & t \le \tau, \\ q_{t-1} - 2q_\tau\gamma_d e^{\gamma_d(t-\tau)}/[1 + e^{\gamma_d(t-\tau)}]^2, & t > \tau. \end{cases} \tag{1}$$

The implementation evaluates the second branch with negative exponential arguments to avoid overflow while preserving the algebra. The growth coefficient is interpreted in the annual recurrence's time convention; a dimensional continuous-time growth coefficient would have different units. An integer peak year is required explicitly. The emissions prior allows continuous peak years, but the treatment of q_τ for a noninteger peak was not recovered.

The 2024 manuscript gives independent truncated-normal priors for growth, peak year, and decline: means $(0.004, 2070, 0.07)$, standard deviations $(0.0075, 25, 0.05)$, and bounds $[0.001, \infty)$, $[2030, 2200]$, and $[0.001, 0.2]$, respectively [2]. These are not the posterior distributions used in the target ensemble. The subsequent inversion uses expert cumulative-emissions information whose complete probability-density prescription was not recovered. Sampling these priors would therefore not reproduce the target emissions distribution.

Recoverable carbon kernels are net primary production $N = N_0[1 + \beta \ln(C/C_0)]$, with $N_0 = 60\,\mathrm{Gt\,C\,yr^{-1}}$ and $C_0 = 280\,\mathrm{ppm}$, and heterotrophic respiration $(a_1 D + a_2 S)Q_{10}^{\Delta T/10}$ [4]. Here D and

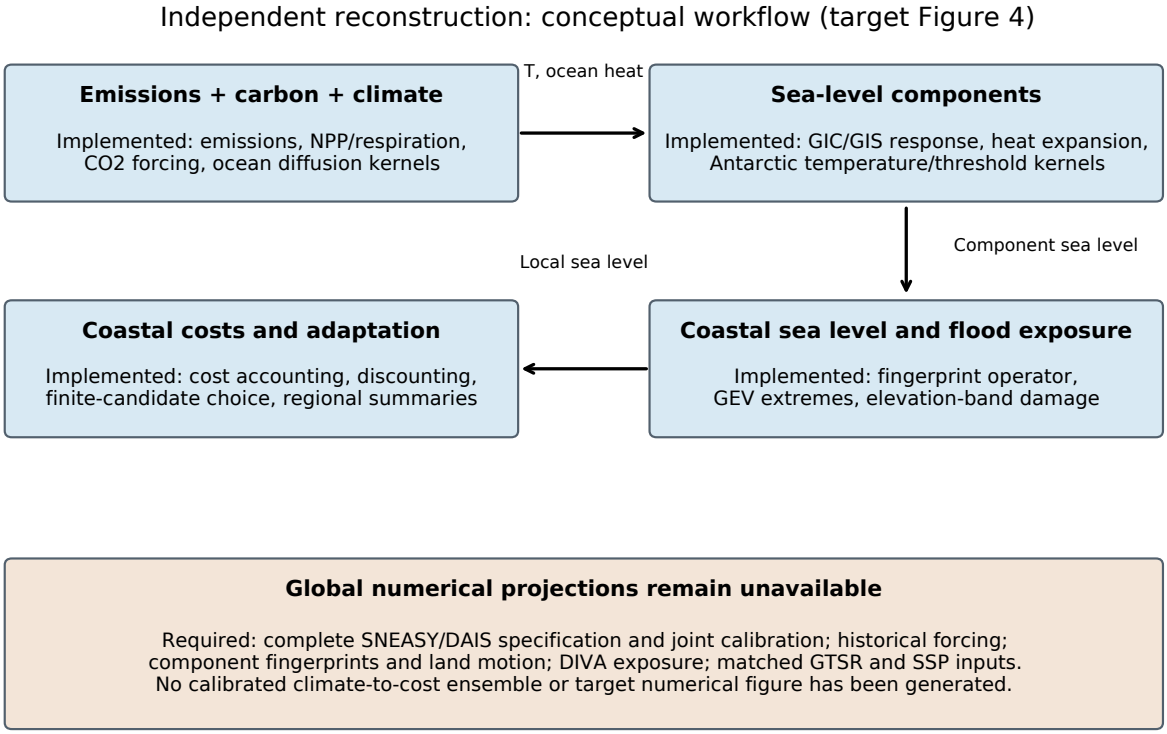


Figure 1: Original conceptual schematic corresponding to target Figure 4. Blue boxes identify implemented component calculations and their physical interfaces. The lower box lists unresolved inputs and model closure. The arrows describe required exchanges, not a claim that a complete coupled ensemble has been executed. This diagram is not counted as a reproduced numerical result.

S are detrital and soil carbon, and a_1, a_2 are turnover rates. The full terrestrial allocation and nonlinear ocean-carbon buffering were not reconstructed. Consequently, these expressions are not used to claim a closed global carbon budget. The implemented CO_2 forcing is $3.7 \log_2(C/C_0)\,\mathrm{W\,m^{-2}}$ [3].

The ocean verification operator solves $\partial_t T = \kappa \partial_{zz} T$ on a one-dimensional column with prescribed surface temperature and zero bottom heat flux. Cell-centred finite volumes use half-cell distance at the upper boundary. Backward Euler time integration ensures that the column heat change equals the top flux evaluated at the new time, to roundoff. Multiplication by volumetric heat capacity converts the tested temperature integral to heat per unit area. This independently chosen discretization is not asserted to match the original SNEASY solver.

3.2 Sea-level components and spatial transformation

The glacier contribution S satisfies the documented reservoir relation

$$\frac{dS}{dt} = \beta (T - T_{\mathrm{eq}})(1 - S/V_0)^n, \qquad T_{\mathrm{eq}} = -0.15\,^{\circ}\mathrm{C}. \tag{2}$$

Temperature is referenced to 1850–1870, V_0 is initial glacier volume expressed as metres of sea-level equivalent, and β has units $\mathrm{m\,K^{-1}\,yr^{-1}}$ [5]. The numerical domain requires $S \le V_0$ for fractional n. An integration step that leaves this domain raises an error; no undocumented volume clipping is introduced.

Greenland volume V, also in metres of sea-level equivalent, follows

$$V_{\mathrm{eq}} = cT + b, \qquad \frac{dV}{dt} = (V_{\mathrm{eq}} - V)(\alpha T + \beta). \tag{3}$$

The temperature reference is 1961–1990, and the volume loss relative to its initial state yields sea-level rise. The rate $\alpha T + \beta$ must be nonnegative. Constant temperature gives an exponential relaxation and therefore an independent verification solution.

The target lineage uses a heat-driven thermal-expansion response, $\Delta S_{\mathrm{TE}} = k_H \Delta H$, not the earlier temperature-relaxation emulator [9, 2]. The coefficient k_H in $\mathrm{m\,J^{-1}}$ remains an explicit input. If the density expansion parameter α_ρ has units $\mathrm{kg\,m^{-3}\,K^{-1}}$, dimensional mass and heat accounting gives $k_H = \alpha_\rho / (A C_p \rho^2)$. This is a dimensional derivation, not recovery of a calibrated numerical coefficient. Supplying a conventional volumetric coefficient in $\mathrm{K^{-1}}$ in that expression would be a unit error.

The documented Antarctic ocean-temperature transformation is

$$T_o = T_f + \frac{x}{1 + e^{-x/a}}, \qquad x = aT + b - T_f, \qquad T_f = -1.8\,^{\circ}\mathrm{C}. \tag{4}$$

An additional threshold-triggered loss rate can be evaluated when its threshold and rate are supplied. Full DAIS dynamics are not implemented: the target requires a particular corrected version, global-sea-level feedback, paleo calibration, and initial state. The axisymmetric geometry and conservation equations in Shaffer[7] are documented in the recovery record, but that standalone model has not been established as an interchangeable target component. Land-water increments likewise remain externally specified.

Local relative sea level is represented by $L_i(t) = \sum_c F_{ic} S_c(t) + v_i(t - t_0)$, where F contains component fingerprints and positive v_i increases relative sea level. This sign convention includes subsidence as positive. The matrix operation is verified on known area-weighted factors; actual fingerprint grids and coast matching are absent. Reference-period routines require complete annual coverage and refuse silent rebasing of an incomplete series.

3.3 Coastal damage and costs

The independently implemented flooding calculation uses a generalized extreme value (GEV) distribution for annual extreme water level and piecewise uniform area density within supplied elevation bands [8]. For water height w and exposed elevation e, property damage is proportional to $\phi(w-e) = (w-e)/(h_0+w-e)$ with $h_0 = 1\,\mathrm{m}$. The integral over an exposed band $[a,b]$, where $b \le w$, is

$$\int_a^b \phi(w-e)\,de = (b-a) - h_0 \ln\left(\frac{h_0+w-a}{h_0+w-b}\right). \tag{5}$$

Multiplication by area density and capital density yields event loss. Mortality uses flooded population, a supplied mortality fraction and value of statistical life. Both contributions are multiplied by $1-r$, where r denotes economic resilience. Integration of this loss against the GEV density produces expected annual loss. The surge integration bound is explicit, and the truncated density is not renormalized. Retreat changes the lowest retained elevation; protection changes the overtopping threshold.

Relocation combines income-related moving cost, mobile-capital transfer cost, and demolition of immobile capital. The documented benchmark mobile share is one quarter; immobile capital is three quarters. Reactive relocation costs five times the proactive counterpart in the target model. Separate kernels calculate inundation with explicit depreciation, wetland loss with a critical rise rate, and quadratic-height seawall construction. Annual maintenance is returned as a separate cash flow. The numerical coefficient in the construction formula must have units consistent with its quadratic height dependence; the quoted historical reference cost is not silently substituted where its unit convention is ambiguous.

For annual payments C_y, the explicit discount operator is

$$\mathrm{PV}_{y_0} = \sum_y \frac{C_y}{(1+r)^{y-y_0}}. \tag{6}$$

A 2010 base year and 4% rate are used in verification. This base-year convention does not resolve the target caption's wording about discounting through/to 2150. The optimizer chooses the least-cost member of a supplied finite candidate menu. It does not construct the original adaptation trajectories, because their physical and socioeconomic inputs are missing.

4 Verification results

All 80 Julia assertions pass across seven groups: emissions/carbon/forcing; ocean conservation and convergence; sea-level limits; spatial accounting and references; extreme levels and damage integration; costs and optimization; and ensemble/regional arithmetic (the sea-level group includes ODE accuracy). These are numerical verification tests, not validation against Earth observations. The tests require no third-party Julia packages. Configuration values are explicitly marked as manufactured in `config/validation.toml`. Table1 is generated directly from the diagnostic files and the independent SciPy checks.

4.1 Printed emissions recurrence and its limit

Figure2 compares the literal recurrence with the continuous logistic decline $q(t)/q_\tau = 2/[1+e^{\gamma_d t}]$. The decrement in Equation1 is the right-endpoint sample of the continuous decline rate. Summing endpoint samples does not exactly integrate that rate. For small step Δt, the residual fraction is

Verification quantity	Calculated value
Greenland finest absolute error (m SLE)	5.398e-10
Ocean finest maximum temperature error (K)	6.759e-06
Ocean maximum heat-balance residual (K m)	1.021e-16
Expected flood loss, SciPy reference (USD/year)	442422.479342
Expected flood loss, Julia relative difference	1.271e-11
Depth integral maximum difference (USD)	9.313e-10
Annual emissions residual at gamma=0.07 (% of peak)	1.750000
Minimum sampled Antarctic ocean transform (degree C)	-2.078464

Table 1: Computed verification metrics. These quantities describe manufactured test cases, not target global projections. The heat-balance residual is reported before multiplication by volumetric heat capacity.

$\gamma_d \Delta t/4$ to leading order. At an annual step and decline rate $0.07\,\mathrm{yr}^{-1}$, the calculated residual is approximately 1.75% of peak emissions. The tested rates 0.01 and $0.2\,\mathrm{yr}^{-1}$ give approximately 0.25% and 5%, respectively. Refining the quadrature step reduces the residual. These are normalized mathematical cases extending 2000 years after an arbitrary peak, not calendar-year emissions projections.

The discrepancy separates two interpretations of the published specification: literal annual accumulation and a continuous logistic trajectory that vanishes asymptotically. No choice was imposed on a global simulation. The effect on temperature and costs cannot be calculated without the missing coupled model.

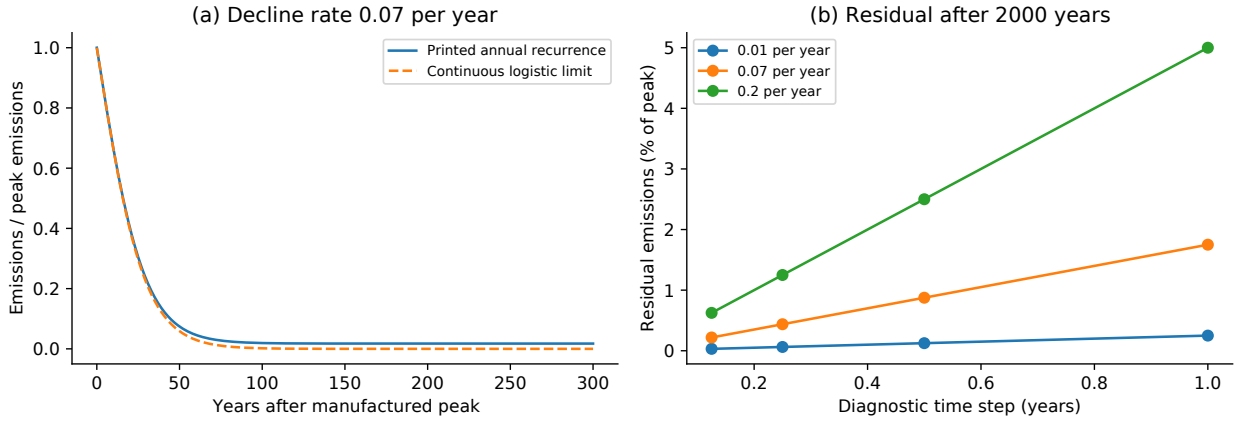


Figure 2: Independent diagnostic of the printed emissions equation. (a) Annual endpoint accumulation for normalized peak emissions and decline $0.07\,\mathrm{yr}^{-1}$, compared with the continuous logistic expression. (b) Remaining emissions after 2000 years for three decline rates and four diagnostic step sizes. The residual decreases with step size; this does not establish which interpretation the original software used.

4.2 ODE and ocean-diffusion convergence

Figure3 tests the Greenland relaxation against its exact constant-temperature solution. The manufactured case has $T = 2\,\mathrm{K}$, $c = -0.2\,\mathrm{m\,K}^{-1}$, $b = 1\,\mathrm{m}$, $\alpha = 0.005\,\mathrm{K}^{-1}\,\mathrm{yr}^{-1}$, $\beta = 0.01\,\mathrm{yr}^{-1}$ and

initial volume 1.2 m. The equilibrium is 0.6 m and the integration lasts 100 years. Fourth-order convergence is obtained when the RK4 step is halved.

The ocean case uses a 1 m column, diffusivity $0.1\,\mathrm{m^2\,s^{-1}}$ and initial temperature $\sin[\pi z/(2L)]$ K. The upper temperature is zero and the bottom flux vanishes. The exact amplitude decays as $\exp[-\kappa(\pi/2L)^2 t]$. Time steps decrease with squared grid spacing, so the first-order temporal and second-order spatial errors converge together at second order in spacing. These deliberately artificial dimensions isolate the numerical operator from Earth-system calibration.

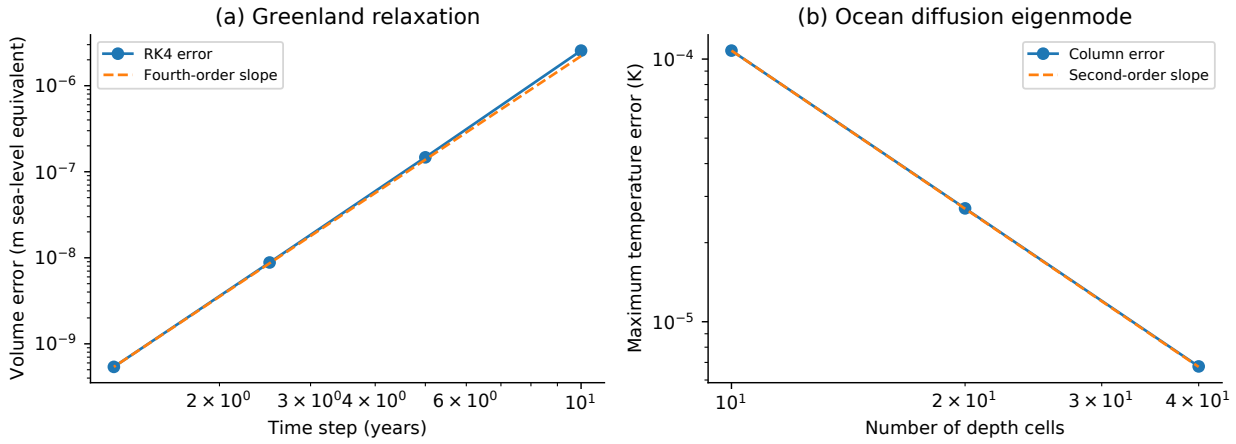


Figure 3: Independent convergence tests. (a) Greenland volume error at 100 years against the exact exponential relaxation; the dashed line has fourth-order slope. (b) Maximum ocean temperature error at 0.2 s for 10, 20 and 40 cells; the dashed line has second-order slope. Time step is 0.2 divided by the square of the cell count, in seconds.

4.3 Flood-loss integration and decision sensitivity

Figure4 uses a manufactured $5\,\mathrm{km^2}$ segment distributed uniformly over 0–5 m elevation. Capital density is one million USD $\mathrm{km^{-2}}$, population density $100\,\mathrm{km^{-2}}$, mortality fraction 0.01, value of life 100,000 USD and resilience 0.2. These values are verification fixtures, not recommended valuations. The GEV parameters are location 1 m, scale 0.3 m and shape 0.1. The outer integral extends from -2 to 20 m. SciPy uses its opposite-sign shape convention and independent adaptive integration for both elevation and surge. This verifies the Julia closed-form depth integral and composite-Simpson expectation.

Table2 quantifies two accounting ambiguities with manufactured costs. A two-period menu has candidate costs $(1, 9)$ and $(6, 2)$ at years 2010 and 2060. Keeping one candidate differs from allowing a switch. This menu has no inherited infrastructure state; it is a test of the decision rule only. Likewise, ten annual unit payments during 2010–2019 differ from paying ten units at either end. A coarse timestep must retain the annual-flow interpretation if it is to reproduce the same present value.

4.4 Antarctic transform and its asserted floor

For $a > 0$ and $x < 0$, the added term in Equation4 is negative. The literal transformation therefore falls below T_f, despite its description as bounded below by freezing [5]. For the test values $a = 1, b = 0$, the minimum is approximately $-2.07846\,^\circ$C. In dimensionless form, $z = x/a$, the

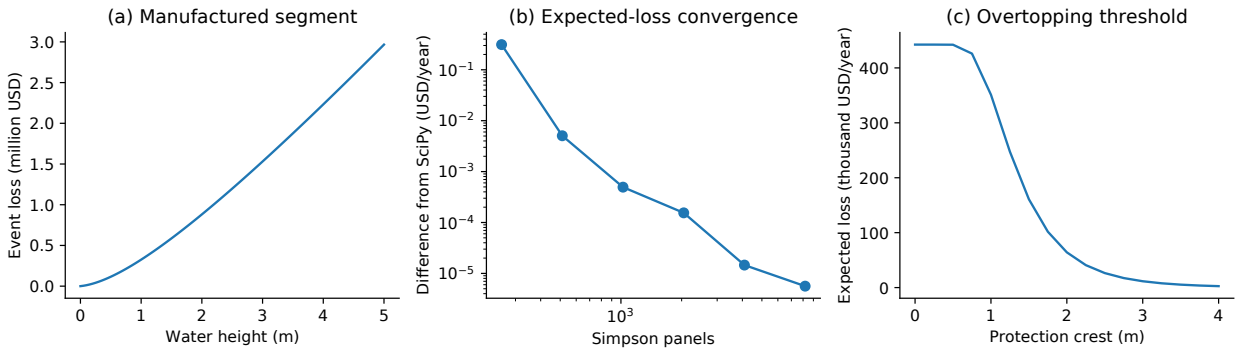


Figure 4: Manufactured coastal damage verification. (a) Event damage increases with water height. (b) Expected annual loss approaches an independent nested SciPy integral as the number of Simpson panels increases. (c) Expected overtopping loss decreases with protection crest. Panel(c) omits construction cost and therefore is not an optimal seawall-height calculation.

Manufactured accounting choice	Present value
fixed strategy pv	2.266414
revisable strategy pv	1.281425
ten annual unit payments pv	8.435332
ten units paid at start pv	10.000000
ten units paid at end pv	7.025867

Table 2: Sensitivity to strategy persistence and payment timing at 4% annual discounting. Units are arbitrary currency; these are not coastal estimates.

extremum satisfies $1 + e^{-z} + ze^{-z} = 0$, giving $z \simeq -1.27846$ and a minimum increment $-0.27846a$. Figure5 shows this algebraic behavior. No freezing clamp was added. Whether the calibrated historical trajectory visits this regime remains unknown.

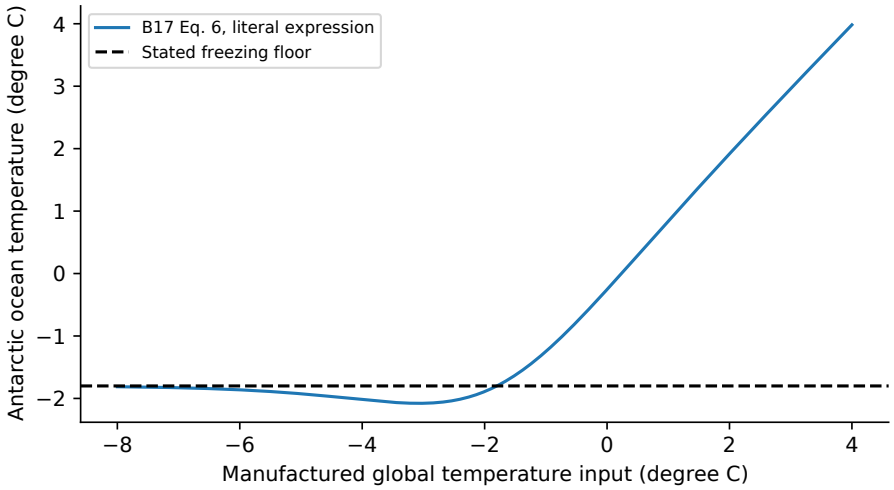


Figure 5: Literal BRICK Antarctic ocean-temperature transformation with manufactured coefficients $a = 1, b = 0$ and stated freezing temperature $-1.8\,^{\circ}\mathrm{C}$. The curve falls below the dashed floor for negative x. This is a specification diagnostic, not an Antarctic projection.

5 Quantitative comparison boundary

The target reports median costs of 1.2, 1.5, 2.1 and 2.6 trillion 2010 USD across the four warming bins, with fifth percentiles 1.0, 1.2, 1.6 and 2.2 trillion USD [1]. The prose supplies 95th percentiles 1.3 and 2.3 trillion USD for the first two bins; unreported numerical endpoints were not digitized or invented. No independent global costs exist here, so neither relative errors nor agreement intervals can be calculated for those targets.

An arithmetic audit of the printed Table 1 is possible. Its four counts sum to 5,000. Among 250 simulations above 2.7 trillion USD, the text gives 125 in the highest bin and 109 in the third, with none in the lowest. The remaining 16 belong to the second bin. Dividing these counts by bin size yields the rounded probabilities in the paper (Table3). This verifies internal consistency of printed summaries; it does not reproduce the modeled probability distribution or establish an independently calculated tail threshold.

Bin	Printed count	Tail count	Calculated%	Printed%
1	202	0	0.0000	0
2	2264	16	0.7067	1
3	2145	109	5.0816	5
4	389	125	32.1337	32

Table 3: Arithmetic audit using published counts only. The second-bin tail count 16 is derived by subtraction. These data never enter the model kernels.

6 Complete figure-by-figure record

Table4 covers all main and supplementary figures and the main table. Unlettered joint and marginal views are listed separately. Every entry's dependencies, source provenance, assumptions, commands, comparison status and failure explanation are also recorded in `docs/FIGURE_STATUS.csv`. For unavailable outputs, the generation command is explicitly unavailable; there is no plotting command that can turn missing model predictions into an independent reproduction. Supplementary Figure 10 is one histogram containing all-member and Antarctic-trigger-conditioned distributions of proactive migrants.

Table 4: Target reproduction status. "Not reproduced" means no independently generated target numerical result.

Figure/view	Quantity	Status
1: global interval plot	Conditional median and 5th-95th cost quantiles	Not reproduced
2: joint scatter	Sea level 2100 versus discounted costs	Not reproduced
2: top marginal	Sea level distribution by warming bin	Not reproduced
2: right marginal	Cost distribution by warming bin	Not reproduced
3: regional bars	Regional cost/GDP minus global cost/GDP	Not reproduced
4: workflow	Model components and exchanges	Conceptual schematic
S1: interval plot	5000 versus 4000 subsample comparison	Not reproduced
S2: interval plot	202 per bin versus full sample	Not reproduced
S3: trajectory plot	GMST median and 90% intervals plus AR6 boxes	Not reproduced
S4: trajectory plot	GMSL median and 66% intervals plus AR6 boxes	Not reproduced
S5: map	Median local sea level 2100	Not reproduced
S6: interval plot	GTSR versus DINAS cost intervals	Not reproduced
S7: joint scatter	Sea level versus costs above 2.7 trillion	Not reproduced
S7: top marginal	Upper-tail sea level distribution	Not reproduced
S7: right marginal	Upper-tail cost distribution	Not reproduced
S8: regional bars	SSP2 regional cost/GDP deviations	Not reproduced
S9: regression plot	Regional SSP income contrast versus cost sensitivity	Not reproduced
S10: histograms	Migration conditional on AIS trigger	Not reproduced
Table1: table	Warming-bin counts and tail probabilities	Not reproduced

Main Figure 1 requires full conditional cost distributions. Figure 2 requires joint sea-level/cost realizations, and Figure 3 additionally requires regional economic aggregation. Supplementary Figures 1 and 2 require controlled subsampling of a legitimate ensemble, not merely a sample-size calculation. Figures 3 and 4 require calibrated time series plus AR6 comparisons. Figure 5 requires spatial fingerprints and site matching. Figure 6 requires matched GTSR and DINAS extreme levels. Figure 7 requires the original type of upper-tail joint distribution. Figures 8 and 9 require paired SSP2/SSP4 economics. Figure 10 requires retained segment migration histories and Antarctic-trigger states. The implemented summary and aggregation routines do not supply these missing physical inputs.

7 Unresolved inputs and implications

The joint parameter distribution is a central missing scientific object. The cited methodology combines paleo Antarctic constraints, modern climate and component-sea-level observations, uncertain emissions, and autoregressive residuals. Complete target likelihood definitions, data vintages, measurement errors, initial states, and residual treatment were not recovered. Reusing published

posterior samples is prohibited here. Independent calibration would be permissible, but it first requires these specifications and the corresponding exogenous observations. Independent draws from older marginal quantiles would discard physically relevant dependence.

The historical DIVA exposure bundle is another independent obstacle. The CIAM supplement reports that its v1.5.5 database was no longer supported and that v1.8 was not public at that time [8]. The EEA provider directory currently reports unavailable software [11]. This does not prove that every DIVA dataset is inaccessible; it establishes that a complete permissible target bundle was not recovered in this investigation. No modern coastline segmentation was substituted for the 12,148 target segments.

Independent GTSR and SSP data records were located [12, 13]. Their existence does not determine which GDP interpretation, scenario vintage, currency conversion, coastal mapping, extreme-value fitting method, or post 2100 extension the target used. The 2022 coastal-uncertainty methods [10] identify six uncertain socioeconomic parameters, but access attempts for their academic supplementary PDF returned challenge or access pages. Those pages were not treated as valid PDFs, and no code repository was used as a workaround. Acquiring large inputs before resolving these interfaces would consume resources without establishing a faithful reconstruction.

The next scientifically defined stages are therefore: recover and version the complete calibration inputs and likelihood; complete and verify SNEASY and the target DAIS variant; recover permissible historical segment and economic inputs; resolve return-period, strategy, and cash-flow conventions; then run small coupled ensembles before scaling to 5,000 members. No result in this report supports a quantitative conclusion about the target's regional cost inequality.

8 Assessment of the research-agent process

The agent recovered much of the mathematical interface from distributed scholarly sources, distinguished versions of a mutable preprint, and implemented several independently checkable numerical components. The analytical tests separate solver accuracy from calibration and data availability. Cross-language flood integration supplies an additional check against transcription or implementation errors in the elevation primitive and GEV convention.

Two interpretation errors were avoided through source comparison. First, the older BRICK temperature-based expansion emulator is not the heat-based component described in the target lineage. Second, a later revision behind the same DOI cannot automatically be treated as the manuscript available before publication. Direct calculation then identified two tensions between printed equations and their prose descriptions. These checks do not reveal the original implementation.

The principal limitation is evidence recovery, not numerical throughput. An agent can write functioning code while failing to identify the scientific experiment if it supplies convenient values for missing inputs. This project does not count such code as a successful reproduction. No observation-based calibration, full carbon closure, complete DAIS integration, or global adaptation ensemble has been demonstrated. The kernel verification establishes a narrower capability: translation of documented equations into testable calculations with explicit domains, units, and limiting cases.

The research log records source access failures, incidental search exposure, interpretation decisions, and verification outcomes. No subagents were used. The work followed Brad Lipovsky's supplied research instructions and academic style guide. The final assessment concerns this bounded execution; it is not a general benchmark of all AI research systems.

9 Reproducibility and conclusions

Run `bash scripts/run_all.sh` from the project root to execute the offline verification, generate numerical diagnostics and figures, and compile this report. Julia uses standard libraries only. Python 3.6.8, NumPy 1.19.5, SciPy 1.5.4 and Matplotlib 3.3.4 were available locally; pdfLaTeX comes from TeXLive 2018. The driver verifies executable availability and imports, requires no installation, and limits numerical libraries to one thread. Calculations run at scheduling priority 19 on a heavily loaded shared server. All artifacts, temporary files and tool caches are directed into the designated project. Source acquisition is a separate optional, hash-checked command. The scientific build does not require network access or downloaded authors' outputs.

The independent calculations verify recoverable response and cost kernels and quantify two written-specification discrepancies. An original conceptual Figure 4 is complete. Main Figures 1–3, supplementary Figures 1–10 and modeled Table 1 remain unreproduced. A printed-table arithmetic agreement is not promoted to numerical-model agreement. The missing calibration and coastal inputs must be recovered before a defensible coupled ensemble or assessment of the target scientific conclusions can be produced.

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[13] IIASA. Shared Socioeconomic Pathways Scenario Database. Provider documentation, accessed 12 October 2026. Multiple GDP interpretations and data vintages require explicit selection before a target reconstruction.